Explicit Diophantine equations for Con(ZFC)

A 61-operation system and a 67-operation polynomial
for the fixed blank computation K215

Research contribution prepared for the ProveIt repository

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Abstract

We give a completely specified polynomial in fifteen strictly positive integer unknowns whose solvability is equivalent, provably in ZFC, to the existence of a finite ZFC refutation. Its negated solvability is therefore equivalent to Con(ZFC). The polynomial has exact degree 155 and an explicit straight-line evaluation with $67 = 39M + 28A$ binary arithmetic operations. An equivalent six-equation system uses $61 = 34M + 27A$ operations. A one-operation tradeoff lowers the polynomial degree to 123. A division-free substitution gives corresponding equations in 41 unrestricted integer unknowns.

The construction specializes the repository's native-window/Pell compiler to the fixed blank-input K215 machine, deleting the numerical input interface. All four coefficient ports are bound to specific integers by closed finite formulas and the pinned 430-row transition table. There is no unspecified universal-machine index. The principal cost of the short arithmetic description is coefficient height: the cell base is $B = 2^{5^{67}}$. The coefficients and the approximately $1.42 \cdot 10^{21}$ local windows are defined exactly, but are not expanded.

This contribution consists of a mathematical proof, explicit arithmetic circuits, source pins, and reproducible finite audits. The new Diophantine bridge has not been checked by Lean or Rocq. The existing K215 endpoint formalizations are identified separately. No global minimality claim and no proof of Con(ZFC) is made.

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1 Result, domain, and the simplicity criterion

The convention throughout the main construction is that all supplied unknowns range over $\mathbb{Z}_{>0}$. Every intermediate expression is an ordinary integer polynomial; intermediate values need not be positive off the zero set. Section 9 removes the positivity convention.

Theorem 1.1 (Fixed consistency equation)**.** *Let the four integers $m, K, M_C, M_F^{\mathrm{src}}$ be those defined in Sections 3–4, with the K215 transition table at the source snapshot stated on the title page. Let P_{67} be the polynomial in Section 2. Then*

$$\mathrm{ZFC} \ \vdash \ \Big[\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{67}(\mathbf{x}) = 0 \ \longleftrightarrow \ \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)\Big]. \tag{1}$$

In particular,

$$\mathrm{ZFC} \ \vdash \ \Big[\forall \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{67}(\mathbf{x}) \neq 0 \ \longleftrightarrow \ \mathrm{Con}(\mathrm{ZFC})\Big]. \tag{2}$$

The same assertions hold for the six-equation system in Section 2, for P_{68}, and, after the explicit substitution in Section 9, over $\mathbb{Z}^{41}$.

Here $\mathrm{Proof}_{\mathrm{ZFC}}$ denotes an ordinary effective finite proof predicate for ZFC. Changing among the standard effective presentations of ZFC changes neither consistency nor the result: the finite proof translations are formalizable in ZFC. The calibration to the exact presentation used by the machine is explained in Section 7.

We minimize the size of a *complete explicit arithmetic description* within the repository's existing compiler framework. Each binary addition, subtraction, or multiplication costs one operation, including multiplication by a fixed numeral. Fixed numerals, reading unknowns, reusing a register, and comparing two computed registers cost zero. Thus the system and single-polynomial figures measure different interfaces. A sum-of-squares row includes all residual subtractions, squares, and additions.

Presentation	Unknowns	Equations	Operations	Degree
Basic system	15 positive	6	$61 = 34M + 27A$	60 maximum
Reduced-coefficient system	15 positive	6	$62 = 34M + 28A$	46 maximum
$P_{67} = 0$	15 positive	1	$67 = 39M + 28A$	155
$P_{68} = 0$	15 positive	1	$68 = 39M + 29A$	123
Basic sum of squares	15 positive	1	78	120
Reduced sum of squares	15 positive	1	79	92
P_{67}, integer substitution	41 integer	1	158	310
P_{68}, integer substitution	41 integer	1	159	246

The smallest complete arithmetic construction obtained here is the 61-operation system. These are upper bounds for displayed circuits, not arithmetic-circuit lower bounds. Fifteen is not a minimum variable count: classical universal Diophantine constructions can use nine positive unknowns, with a very different degree and description-size tradeoff [3]. No effective nine-variable specialization is claimed to have been emitted in this archive.

2 The polynomial and the six equations

Supply exactly these fifteen positive integers:

$$\mathbf{x} = (J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma). \tag{3}$$

The symbol K below is a fixed coefficient. The lower-case k is a polynomial abbreviation, not a sixteenth unknown. Define

$$\begin{aligned} &q = mJ + 1, \quad C = q - F - \alpha, && X = wq, && Y = sq^3, \quad E = XY, \\ &k = \eta + \zeta, \quad c = kY + \eta, && a = E + Y, \\ &H = 4a + 3, \quad \Delta = a^2 + H = (a+2)^2 - 1, && D = X + ac + \gamma H. \end{aligned} \tag{4}$$

Next put

$$\begin{aligned} G &= (q-1)(q-F) + \alpha = q^2 - C - qF, \\ R &= G(q^2-1) + (M_C + qM_F^{\mathrm{src}})J, \\ L_0 &= EkY = XY^2k, \qquad V = c(Tf-1) - Rf^2, \qquad S = (i\Delta c^2)^2. \end{aligned} \tag{5}$$

Finally form the six factors

$$\begin{aligned} N_0 &= \tau^2 - L_0(L_0 + k), \\ N_1 &= D^2 - \Delta c^2, \\ N_A &= S(V^2 - y^2) + y^2, \\ N_K &= k - hE - R, \\ N_T &= (K + w)C + q - F - z(q-1), \\ N_S &= \Delta f^2 - S. \end{aligned} \tag{6}$$

The requested single equation is

$$\boxed{P_{67} = N_0N_1N_AN_KN_TN_S - \Delta = 0.} \tag{7}$$

Equivalently, one may use the system

$$\boxed{N_0 = N_1 = N_A = N_K = N_T = 1, \qquad N_S = \Delta.} \tag{8}$$

Abbreviations in (4)–(6) are to be substituted, not imposed as extra equations with extra unknowns. All powers occurring in this polynomial have fixed exponents. The very large exponents used later to define the fixed coefficients are also fixed integers.

For the degree-123 alternative, replace only N_A by

$$N'_A = \Delta(f^2-1)(V^2 - y^2) + y^2, \qquad P_{68} = N_0N_1N'_AN_KN_TN_S - \Delta. \tag{9}$$

Section 8 proves that this replacement preserves the zero set; it is not an off-zero substitution of equal expressions. The archive's JSON packets contain every gate, fixed-numeral port, witness port, output register, and operation ledger. Appendix A prints the complete 67-gate source.

3 The fixed machine and its local computation language

3.1 Pinned K215 semantics

The source table is `Computability/BusyBeaver/Research/ZFC/candidate-k215/k215.tm`. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

There are 215 working states $0, \ldots, 214$, halt state 215, two tape symbols $0, 1$, start state 0, and an initially blank two-sided tape. The 430 rows specify a write, a left/right move, and a next state. The first transition is

$$(0, 0) \longmapsto (1, +1, 93). \tag{10}$$

The unique transition into the halt state is $(68, 1) \mapsto (1, +1, 215)$. For tableau padding only, complete the table by $(215, b) \mapsto (b, 0, 215)$. This preserves the write and movement on the transition entering the halt state. No prefix or auxiliary machine is added. The verifier independently parses the export, the literal Lean table `KDense215.lean`, and the literal Rocq table `Table215.v`, and compares all 430 rows. A finite bootstrap check is supplementary evidence; it is not the unbounded halting theorem.

3.2 A 439-symbol cell alphabet

Let $\mathcal{A}$ be the following alphabet, with the indicated integer codes:

Code	Meaning
b, $b \in \{0, 1\}$	ordinary tape symbol b, no head
$2(q_0 + 1) + b$, $0 \le q_0 \le 215$	symbol b, head in state q_0
$434, 435, 436$	initialization symbols L, Q, R
437	vertical boundary V_{bd}
438	horizontal boundary H_{bd}

The payloads of L, Q, R are respectively $(0, \varnothing), (0, 0), (0, \varnothing)$. Their phases are L, Q, R. Ordinary symbols have no phase. V_{bd} has boundary bits $(v, h_b) = (1, 0)$, H_{bd} has $(0, 1)$, and all interior symbols have $(0, 0)$. The exact local predicate is applied to a 3-by-3 window, in row-major order, with time increasing from the top row to the bottom row. It inspects its center, left, right, previous-time, and next-time symbols; the four corners are unrestricted. Require:

1. The center's v-bit equals its next-time neighbor's v-bit. The center and right neighbor are not both V_{bd}; the center and next-time neighbor are not both H_{bd}. No symmetric left/center or previous-time/center test is added to this center predicate.
2. If the center and its right neighbor are both nonvertical-boundary symbols, their h_b-bits agree.
3. At an interior center, a phase is present exactly when the previous-time symbol is H_{bd}. At such a center, if the right neighbor is nonvertical-boundary, the ordered pair of center/right phases belongs to LL, LQ, QR, RR. No left phase-pair test is added. An initialization cell next to a left boundary is L, and one next to a right boundary is R.
4. A head with H_{bd} as its next-time neighbor must be in state 215. A head must not move through an adjacent vertical boundary.
5. If the next-time center is not H_{bd}, its payload is the deterministic three-neighbor update. A current center head writes its tape symbol; otherwise that tape symbol stays. A new head can arrive from the left moving right, the center staying, or the right moving left. Multiple arrivals are forbidden. A vertical boundary is treated as blank and headless. A horizontal-boundary neighbor cannot supply an interior update.

At a boundary center only the boundary conditions apply. The full, executable, unambiguous predicate is `local_valid` in `code/blank_helical.py`; the preceding clauses specify the same finite relation.

3.3 Start and an impossible distinguished window

Two locally valid windows, in row-major code order, are

$$S_0 = \begin{pmatrix} 438 & 438 & 438 \\ 434 & 435 & 436 \\ 0 & 1 & 188 \end{pmatrix}, \qquad E_0 = \begin{pmatrix} 0 & 438 & 0 \\ 0 & 437 & 438 \\ 0 & 437 & 0 \end{pmatrix}. \tag{11}$$

Every initialized Q in a globally valid tableau has window S_0: the actual first transition (10) forces the next row. Conversely, S_0's center is an initialized Q. The window E_0 satisfies the center predicate, but cannot occur globally. Its previous-time center is H_{bd} with vertical bit zero, while that symbol's next-time neighbor is V_{bd} with vertical bit one. The predicate at that H_{bd} contradicts vertical-bit propagation. We reserve the compiler's forbidden selector 1 for this window. No genuine computation is removed by forbidding it.

Lemma 3.1 (Marked periodic tableau)**.** *There exists a doubly periodic globally valid $\mathcal{A}$-tableau containing S_0 if and only if K215 halts on blank input. For the forward direction, global uniqueness of S_0 is unnecessary. For the reverse direction, its width and height can be padded independently to arbitrary sufficiently large integers, in particular to sufficiently large powers of five.*

Proof. Start with a periodic tableau containing S_0. On the initialization row containing its Q, the phase graph $L \to L, Q;\ Q \to R;\ R \to R$ cannot form a cycle through Q. Horizontal periodicity therefore supplies vertical boundaries on both sides. Choose the nearest ones. Vertical-bit propagation makes them the sides of a finite strip. The horizontal boundary through the previous-time neighbor extends across the strip, and temporal periodicity supplies a next boundary. Choose the first one. The first interior row is $L^u Q R^v$ with $u, v \geq 1$. It has one head in state zero and all tape cells blank. Induction gives the exact deterministic run in the strip; the head cannot escape. Its last interior row has its unique head in state 215. Extending the strip by blank tape recovers the actual blank run.

Conversely take a finite halting run. Choose a width $w_0 = u + v + 2$ with positive margins containing every visited tape cell. Make column zero vertical boundary, the first time row horizontal boundary away from column zero, the next row $L^u Q R^v$, then the run followed by enough absorbing halt rows. Call this rectangle G_0, of height H_0. Width and height can be increased independently. Extend it to the plane by

$$T_0(x, t) = G_0\big(x \bmod w_0,\ (t + \lfloor x/w_0 \rfloor) \bmod H_0\big). \tag{12}$$

The vertical-boundary column permits the one-row time displacement at each horizontal seam: its payload is absent, it is treated as blank in updates, and horizontal boundary-bit propagation stops there. All clauses remain valid. The periods are $(w_0, -1)$ and $(0, H_0)$. The resulting canonical quotient has exactly one S_0, and no E_0. □

3.4 Window lift, ordering, and exact cardinality

Let $\mathcal{W} \subset \mathcal{A}^9$ be all center-valid windows. Order them as

$$W_0 = S_0, \quad W_1 = E_0, \quad W_2, \ldots, W_{\kappa-1} \text{ in lexicographic row-major order after these exclusions.} \tag{13}$$

This is a finite, unconditional definition. It performs no halting test. The exact cardinality is

$$\kappa = 38\,256\,607\,897 \cdot 439^4 = 1\,420\,903\,358\,357\,108\,792\,377. \tag{14}$$

Appendix B gives the cross count and its independent finite check. Use the two-column horizontal overlap and the two-row temporal overlap as the ternary relation between a window and its two

neighbors. For any cyclic word with offsets $0, -1, -h_0$, the overlaps identify every entry of every window with the appropriate center symbol, including when some offsets coincide. Thus they reconstruct an actual periodic plane. Conversely actual windows satisfy the overlaps. For the rectangle in the preceding proof, put $N = w_0 H_0$, let x_S be the initial head column, and use

$$j = -(x - x_S) - w_0(t-1) \pmod{N}. \tag{15}$$

The right and next-time offsets are $-1, -w_0$, and S_0 is at index zero. This supplies precisely the cyclic interface of the native compiler.

4 Four completely specified coefficients

This section binds every fixed-numeral port in (4)–(7). The different symbols separate compiler geometry from the Pell variables: $a_0 = 439$ and κ count symbols and windows; neither is the polynomial's a or k.

4.1 Finite layout

Let ℓ be the least integer at least two such that $2^\ell > \kappa + 1$, and set $A_0 = 2^\ell$. Define

$$\begin{aligned}
p_0 &= \max\left(0, \left\lceil \frac{\kappa - 2\ell - 21a_0 - 4}{2} \right\rceil\right), & j_{\max} &= 9a_0 + 4 + p_0, \\
\mu &= A_0 - 2 + \sum_{j=1}^{j_{\max}} A_0^j, & m_0 &= 2\ell + 30a_0 + 8 + 2p_0, \\
n_{\mathrm{pos}} &= m_0 + 1, & n_{\mathrm{dum}} &= n_{\mathrm{pos}} - \kappa - 9a_0 - 4, \\
e_* &= \kappa + 12a_0, & M_0 &= \kappa + 15a_0 + n_{\mathrm{dum}}.
\end{aligned} \tag{16}$$

For $r, c \in \{0,1,2\}$ and $u \in \{0, \ldots, a_0 - 1\}$, put

$$e(r,c,u) = \kappa + 3a_0 + (2-r)3a_0 + (2-c)a_0 + u, \qquad j(r,c,u) = 1 + (3r+c)a_0 + u. \tag{17}$$

The finite set of allowed native positions is the disjoint union

$$\begin{aligned}
\mathcal{E} = {} & \{0, \ldots, \kappa - 1\} \cup \{e(r,c,u)\} \\
& \cup \{e_*, \ldots, e_* + n_{\mathrm{dum}} - 1\} \cup \{M_0, 3M_0, 9M_0, 27M_0\}.
\end{aligned} \tag{18}$$

Its cardinality is n_{pos}. Define nonnegative integer clause coefficients c_e, for $e \in \mathcal{E}$, as follows:

$$\begin{aligned}
c_i &= 1 + \sum_{r,c=0}^{2} A_0^{j(r,c,W_i[3r+c])} + \sum_{t=1}^{4} A_0^{9a_0+t} && (0 \le i < \kappa), \\
c_{e(r,c,u)} &= A_0^{j(r,c,u)}, \qquad c_{3^{t-1}M_0} = A_0^{9a_0+t} && (1 \le t \le 4), \\
c_e &= 0 && \text{at a dummy position.}
\end{aligned} \tag{19}$$

Here $W_i[3r+c]$ is an integer between zero and 438, determined by (13). Let

$$\begin{aligned}
E_{\max} &= 27M_0, \\
H_C &= E_{\max} + 24M_0 + 3a_0 + 1, \\
T_1 &= H_C + 2E_{\max} + a_0 + 1, \\
T_2 &= T_1 + 4E_{\max} + 2.
\end{aligned} \tag{20}$$

Let L be the least power of five at least $T_2 + 2E_{\max} + 2$. For a positive integer v, write $\mathrm{bits}(v) = \lfloor \log_2 v \rfloor + 1$. Put

$$\begin{aligned} U_0 &= 4(n_{\mathrm{pos}} + 2)\big(28(\kappa + 9a_0 + 4) + 6\big) + 8, \\ b_{\min} &= \max\big\{\ell(j_{\max} + 1) + 3,\ \ell(9a_0 + 4) + \mathrm{bits}(U_0) + 1,\ 4\big\}. \end{aligned} \tag{21}$$

Let b be the least power of five at least $b_{\min}$, and define

$$\rho = 2^b, \qquad d = bL, \qquad B = \rho^L = 2^d. \tag{22}$$

These definitions contain only specified finite integer operations, finite sums, and bounded finite enumeration.

4.2 The masks and transport coefficients

Define

$$\begin{aligned} D_C &= \rho^{3a_0} + \rho^{H_C + a_0} + \rho^{8M_0} + \rho^{24M_0} + \sum_{e \in \mathcal{E}} c_e\big(\rho^{T_1 - e} + \rho^{T_2 - e}\big), \\ D_R &= \rho^{H_C}, \end{aligned} \tag{23}$$

and

$$\begin{aligned} M_F^0 = \mu(\rho^{T_1} + \rho^{T_2}) &+ \sum_{\substack{0 \le r \le 1 \\ 0 \le c \le 2 \\ 0 \le u < a_0}} \rho^{e(r,c,u)} \\ &+ \sum_{\substack{0 \le r \le 2 \\ 0 \le c \le 1 \\ 0 \le u < a_0}} \rho^{H_C + e(r,c,u)} + \rho^{9M_0} + \rho^{27M_0}, \\ M_C = B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{e_*}, &\qquad M_F = M_F^0 + 4. \end{aligned} \tag{24}$$

The four coefficients of the delivered polynomial are exactly

$$\boxed{\ m = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F^{\mathrm{src}} = M_F + B - 1. \ } \tag{25}$$

In particular M_F^{src} is not the one-cell mask M_F. Applying $M_F < B - 1$ directly to M_F^{src} would be incorrect.

4.3 Concrete scalar values and coefficient size

The layout formulas yield

$$\begin{aligned} \ell &= 71, & n_{\mathrm{dum}} &= 1, \\ p_0 &= 710451679178554391506, & j_{\max} &= 710451679178554395461, \\ m_0 &= 1420903358357108796332, & M_0 &= 1420903358357108798963, \\ E_{\max} &= 38364390675641937572001, & H_C &= 72466071276212548748431, \\ T_1 &= 149194852627496423892873, & T_2 &= 302652415330064174180879. \end{aligned}$$

Moreover

$$b = 5^{33}, \qquad L = 5^{34}, \qquad d = 5^{67} = 67762635780344027125465800054371356964111328125. \tag{26}$$

Thus $m = B - 1$ alone has exactly 5^{67} binary digits. The archive specifies the exact integers by the finite formulas above; it does not contain their infeasible binary or decimal expansions. It also does not enumerate all κ windows. This is a short coefficient *description*, not a small coefficient-height result.

4.4 Compiler invariants and the fixed modulo-five unit

The clause padding gives $\mathrm{pc}(\mu) = \ell + j_{\max} - 1$ and $m_0 = 2\,\mathrm{pc}(\mu) + 12a_0 + 2$. Consequently the masks obey

$$\begin{aligned} 0 < M_C, M_F < B - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_F \equiv 4 \pmod 8, \\ \mathrm{pc}(M_C) = d - m_0 - 1, \qquad \mathrm{pc}(M_F) = m_0 + 1, \qquad \mathrm{pc}(M_C) + \mathrm{pc}(M_F) = d. \end{aligned} \tag{27}$$

The unused high radix positions make the range inequalities strict. The low selector zero is permitted; selector one is forbidden. The first dummy position has digits $0, 1, 2, 3$ available, and the other native positions have Boolean digits. The conservative bounds

$$\sum_{e \in \mathcal{E}} c_e \le 14(\kappa + 9a_0 + 4) A_0^{9a_0+4}, \qquad \mu < A_0^{j_{\max}+1} \tag{28}$$

and (21) imply

$$\rho \ge \max\left\{ 4(n_{\mathrm{pos}} + 2)\left(2\sum_e c_e + 6\right) + 8,\ 2\mu + 4,\ 16 \right\}. \tag{29}$$

This is the no-carry bound used in the native compiler. The change in the second band in (20) removes a conditional coefficient repair from the older construction. Since b is a power of five, $\rho \equiv 2 \pmod 5$, and $T_2 - T_1 \equiv 2 \pmod 4$. Each paired term in the sum in (23) therefore cancels modulo five. Since $a_0 = 439 \equiv 3 \pmod 4$,

$$2D_C - D_R \equiv 2\big(2 + 3\rho^{H_C} + 1 + 1\big) - \rho^{H_C} \equiv 3 \pmod 5. \tag{30}$$

No uncomputed test or optional high monomial remains in the recipe. The larger band spacing preserves $T_2 - T_1 > 2E_{\max}$, $L - (T_2 - T_1) > E_{\max}$, and $T_2 + E_{\max} < L$, the support inequalities used by the compiler.

5 Arithmetic soundness: every zero recovers a computation

5.1 Unit factors, including the exceptional boundary

Write $A = a + 2$. Positive inputs give $\Delta = A^2 - 1 > 0$ before any equation is imposed. Since $S = \Delta^2 i^2 c^4$,

$$N_S = \Delta N_s, \qquad N_s = f^2 - \Delta i^2 c^4. \tag{31}$$

Equation (7) is consequently equivalent to

$$N_0 N_1 N_A N_K N_T N_s = 1. \tag{32}$$

Every factor is an integer unit. The norm $N_0 = -1$ is impossible. Indeed, for any integer $u \ge 2$,

$$x^2 - u(u+1)v^2 = -1, \quad v > 0 \tag{33}$$

has no integer solution. Choose a minimal v; then $uv < |x| < \sqrt{u(u+1)}v$, and multiplication by the norm-one unit $2u + 1 - 2\sqrt{u(u+1)}$ produces a solution with

$$0 < v' = (2u+1)v - 2|x| < v. \tag{34}$$

This is a contradiction. Apply this with $u = XY^2$. Also $\Delta \equiv 0$ or $3 \pmod 4$, so neither N_1 nor N_s can be -1. Since S is a square, N_A is congruent modulo four to y^2 or V^2, and cannot be -1. Thus

$$N_0 = N_1 = N_A = N_s = 1, \qquad N_K = \varepsilon, \quad N_T = \nu, \quad \varepsilon, \nu \in \{-1, 1\}, \quad \varepsilon\nu = 1. \tag{35}$$

It is essential not to assume $C > 0$ yet. Since

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \le 0 \tag{36}$$

and $K + w \ge 2$, either transport unit forces $C \ge 0$. The boundary $C = 0, F = 2, z = 1, \alpha = q-2$ does give $N_T = -1$; it cannot be discarded by the transport factor alone. Nevertheless $q - F = C + \alpha > 0$, and

$$\begin{aligned} q \le G = q\alpha + (q-1)C \le q^2 - q, \\ 0 < (M_C + qM_F^{\mathrm{src}})J < 2q^2 - q - 1. \end{aligned} \tag{37}$$

Using $q \ge B \ge 16$, we obtain

$$\begin{aligned} q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ R - 2 > 3q + 1, \qquad E \ge q^4 > R + 2, \qquad a = E + Y > R + 2. \end{aligned} \tag{38}$$

These bounds are weaker than a bound sometimes used in the older variable-input proof. They suffice here because the recovered auxiliary target will be exactly R.

5.2 The deleted auxiliary coordinates are positive

The strong norm gives

$$f^2 = 1 + \Delta i^2 c^4, \qquad S = \Delta(f^2 - 1), \qquad f > c^2 > 2c, \qquad R < a < \Delta. \tag{39}$$

Therefore

$$S - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c \ge 1 + \Delta(i^2c^4 - 1) - c > 0. \tag{40}$$

The auxiliary equation is

$$SV^2 - (S-1)y^2 = 1. \tag{41}$$

It excludes $V = 0$. For $v = |V| > 1$, it implies $y \ge v + 1$ and $v^2 - 1 = (S-1)(y^2 - v^2) \ge (S-1)(2v+1)$, hence $v \ge 2S - 1$. But $T > 0$ gives $V > -Rf^2 - c > -S$, excluding negative V of that size. The cases $V = \pm 1$ are excluded by $V \equiv -c \pmod f$ and $f > 2c,\ c > 2$. Hence $V > 0$. Restore, as mathematical expressions only,

$$o = cT - Rf, \qquad j = Tf - R\Delta i^2 c^3 - 1. \tag{42}$$

They are integers and obey

$$of = V + c > 0, \qquad jc = V + R > 0, \qquad V = of - c = jc - R. \tag{43}$$

Thus $o, j > 0$, before any rank theorem or computation decoding. No division or extra witness has been inserted in the polynomial.

5.3 Pell rank and the remaining signs

For an integer $U > 1$, define the standard Pell sequences by

$$(U + \sqrt{U^2 - 1})^n = \chi_U(n) + \psi_U(n)\sqrt{U^2 - 1}, \qquad n \ge 0. \tag{44}$$

We use the established Pell facts from the repository's generic strong-rank, signed step-down, and half-parameter proofs [4, 5, 6]. These are mathematical dependencies, supplied in the archive; they are not consequences of the finite program checks. The first norm, with $P_0 = 2XY^2 + 1$, has fundamental unit $(P_0, 2)$, and consequently

$$\tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \quad n \ge 1. \tag{45}$$

The positive main root similarly gives

$$D = \chi_A(p), \qquad c = \psi_A(p), \qquad p \geq 1. \tag{46}$$

The strong-rank lemma says that, if $c > A\Delta^2$ and a positive ordinary coefficient i_0 satisfies $(i_0c^2)^2 = \Delta(f^2-1)$, then

$$f = \chi_A(m_1), \qquad c \mid m_1, \qquad m_1 \geq c. \tag{47}$$

The signed auxiliary step-down lemma, applied with $m_1 > 2p$, the auxiliary norm (41), $V = of - c = jc - R > 0$, and $0 < p, R < c/2$, gives $R \equiv \pm p \pmod{c}$, hence $p = R$. In this application its auxiliary coefficient is $S = (i_0c^2)^2$. The full congruence argument and its hypotheses are recorded in `proofs/pell_lemma_audit.md`.

Here are the checks of all the size hypotheses. Since $P_0 \equiv 1 \pmod{E}$, the first recurrence gives

$$2n \equiv R + \varepsilon \pmod{E}, \qquad n \geq (R-1)/2 \geq 24. \tag{48}$$

The inequalities $P_0 > A$ and $c > k$ imply $p > n$ by Pell monotonicity. Standard growth then gives $c > A\Delta^2$ and $c > 2p$. Also $k = R + \varepsilon + hE > 2R$, so $c > k > 2R$. Use $i_0 = \Delta i$ in (47); its premise follows from (39). Then $m_1 \geq c > 2p$. Equations (41)–(43) verify the auxiliary hypotheses. We conclude

$$p = R. \tag{49}$$

Write $2n = R + \varepsilon + vE$. The least positive residue in (48) gives $v \geq 0$. If $v \geq 1$, (38) gives $2n > 2R$, contradicting $n < p = R$. Hence $2n = R + \varepsilon$. If $\varepsilon = -1$, then $p = 2n + 1$. Let $Q_0 = \chi_A(2) = 2A^2 - 1 > P_0$. Pell duplication yields

$$\psi_A(2n) = 2A\psi_{Q_0}(n) \geq 2A\psi_{P_0}(n) = Ak. \tag{50}$$

Since $A > Y + 1$, this would give $c = \psi_A(2n+1) > k(Y+1)$. It contradicts

$$kY < c < k(Y+1), \tag{51}$$

which follows directly from the two positive slacks η, ζ. Therefore $\varepsilon = \nu = 1$. The transport factor now excludes $C = 0$, so $C > 0$. This proves the equivalence of (7) and the individual equations (8).

5.4 Power extraction before mask decoding

We explain why the asymmetric scale $X = wq$ suffices. Let $r_0 = (R-1)/2$, now an integer, and put $\xi = (X+1)^{2r_0}/X^{r_0}$. The standard ratio estimates for (45)–(46), using $P_0 > A,\ 6XY^2 > a,\ X \geq q \geq 16,\ Y \geq q^3$, give

$$\xi < \frac{2c}{k} < 2Y + 2, \qquad Y > X^{r_0}/3, \qquad a > X^{r_0+1}/3. \tag{52}$$

These lower estimates do not require $q^3 \mid X$ or the older stronger upper bound on R; see [8], Section 3. In particular $X < a$ and

$$2^R = 2 \cdot 4^{r_0} < X^{r_0+1}/3 < a. \tag{53}$$

The Pell recurrence for $\chi_A(t) - a\psi_A(t)$ is congruent modulo $H = 4a+3$ to 2^t. Thus $D = X + ac + \gamma H$ implies $X \equiv 2^R \pmod{H}$. Both positive representatives are below $a < H$, proving

$$X = 2^R. \tag{54}$$

Because $q \mid X$, $q = 2^t$ for an integer $t > 0$. The bound $R > 3q$ now implies $q^3 \mid X$. Thus one may restore the symmetric scale $\widetilde{w} = X/q^3$ in the proof and apply the full half-binomial kernel [7]. Its hypotheses $q \geq 16,\ 3q + 1 \leq R < q^4$ hold by (38). It gives

$$Y = \frac{1}{2}\left\lfloor \frac{(X+1)^{R-1}}{X^{(R-1)/2}} \right\rfloor,$$
$$\nu_2(Y) = \mathrm{pc}(R) - 2. \tag{55}$$

This uses the central-binomial valuation $\nu_2\binom{2r_0}{r_0} = \mathrm{pc}(r_0)$; all the remaining terms in the binomial floor are divisible by X. As $q^3 \mid Y$, (55) gives $\mathrm{pc}(R) \geq 3t + 2$. Finally $B - 1 = 2^d - 1$ divides $q - 1 = 2^t - 1$. The order of two modulo $2^d - 1$ is d, so $t = dN$ for some $N \geq 1$. Thus

$$q = B^N, \qquad J = 1 + B + \cdots + B^{N-1}. \tag{56}$$

5.5 Inverse packing and synchronization

Let

$$S' = (C - 1) + qF, \qquad T' = (M_C J + 1) + q(M_F J - 1). \tag{57}$$

The source's shifted mask coefficient gives the exact identity

$$R = (q^2 - S')(q^2 - 1) + T'. \tag{58}$$

The now established $C, F > 0$ and $C + F < q$ imply $0 < S' < q^2$. Equation (27) and the repunit (56) give

$$0 < T' < q^2 - 1, \qquad \mathrm{pc}(T') = dN + 2. \tag{59}$$

Adding one to the low block adds one population bit; subtracting one from the high block with exactly two trailing zeros also adds one bit. The elementary inverse-population lemma states that for $\Lambda = 2^n,\ 1 \leq S_1 \leq \Lambda,\ 0 < T_1' < \Lambda - 1$,

$$\mathrm{pc}\big((\Lambda - S_1)(\Lambda - 1) + T_1'\big) \leq n + \mathrm{pc}(T_1'), \tag{60}$$

with equality exactly when $S_1 < \Lambda$ and $S_1 \,\&\, T_1' = 0$. Splitting into its two n-bit blocks proves the assertion: when $S_1 + T_1' < \Lambda$, equality means no addition carry; when $S_1 + T_1' \geq \Lambda$, the outgoing carry and discarded leading bit make the inequality strict. The boundary $S_1 = \Lambda$ is also strict. A detailed proof is included in [12]. Apply (60) to (58). The lower bound $\mathrm{pc}(R) \geq 3dN + 2$ forces equality, and therefore

$$(C - 1) \,\&\, (M_C J + 1) = 0, \qquad F \,\&\, (M_F J - 1) = 0. \tag{61}$$

The first condition restores Start at the origin with no carry, forbids selector one in every cell, and restricts native digits to the positions in (18). It permits additional Start occurrences; their exclusion is unnecessary.

The remaining compiler argument is the native coefficient proof [11, 12], with the layout and masks specified above. We record its relevant content to make the change of local language explicit. The no-carry bound (29) keeps every unshifted coefficient at most $\rho/4 - 2$. In a binary rotation by an arbitrary exponent r, write $r = bt_0 + \epsilon_0$, $0 \leq \epsilon_0 < b$. The extra upper bit at the dummy position means that the possible rotated support is $(\mathcal{E} \cup \{e_* + 1\}) + t_0$ modulo L. For $\epsilon_0 \leq b - 2$, a rotated digit is at most $3\rho/4$; for $\epsilon_0 = b - 1$, it is at most $\rho/2 + 1$. The sum with the unshifted field still has no carry. Let $\widehat{F}$ be the sum of the two native convolutions and that cyclic temporal rotation, represented in the N-cell word. These bounds give $0 < \widehat{F} < q - 1$. The already proved equation $N_T = 1$, together with $X = wq \equiv w \pmod{q-1}$, gives $F \equiv \widehat{F}$

(mod $q-1$). Since $0 < F < q$, the two representatives are equal: $F = \widehat{F}$. Thus the field-mask tests in (61) apply to the actual carry-free field before any clause decoding is used.

The two distant clause bands cannot both meet that rotated support. An uncontaminated band enforces the selector/copy/anchor clauses: at occupied cells it specifies one window, its matching payload-copy bits, and its four anchors. At the marked origin the anchor tests require two distinct rotated digits to be odd. For $1 \le \epsilon_0 \le b-2$ all rotated digits are even. For $\epsilon_0 = b-1$ only the single upper-dummy spill position can be odd. Both cases are impossible. Thus $\epsilon_0 = 0$. The anchor pair differences then force $t_0 = 0 \pmod{L}$. The rotation is a whole-cell shift B^{h_0}. The native occupancy propagation from Start and the copy tests enforce the horizontal and temporal window overlaps. The temporal overlap of Start's horizontal-boundary row and its initialization row excludes $h_0 = 0 \pmod{N}$. No unique-End or unique-Start theorem is used here.

The transport factor used here really gives the required congruence:

$$(K+w)C - F = (z-1)(q-1), \qquad X = wq \equiv w \pmod{q-1}. \tag{62}$$

If a positive quotient for the unsheared form is wanted,

$$(K+X)C = F + (z-1+wC)(q-1), \qquad z-1+wC > 0. \tag{63}$$

Therefore the supplied F is the actual carry-free field, and (61) decodes a marked periodic tableau for the local language of Section 3. The cyclic word pulls back along $(x,t) \mapsto -x - h_0 t \pmod{N}$, with periods $(N,0),(0,N)$; neither $h_0 \mid N$ nor coprimality is needed. Lemma 3.1 proves that K215 halts. This completes arithmetic soundness for every positive zero, including the initially possible $C = 0$ branch.

6 Completeness at the actual packed index

Suppose K215 has a finite blank halting run. We construct positive unknowns without assuming any bound on the length of that run. By Lemma 3.1, choose independent sufficiently large powers of five w_0, H_0, with $H_0 \ge 25$,

$$N = w_0 H_0 > 25d, \qquad q = B^N. \tag{64}$$

Use the canonical cyclic word with one Start, no forbidden window, and all dummy bits initially zero. Let C_0 be its native word and F_0 its actual field for temporal shift B^{w_0}. Let R_0 be the packed rank (5) evaluated at these C_0, F_0. The source and mask proofs in [12] show that (61) holds. In particular the origin's extra field tests hold: the temporal predecessor of the unique Start is not Start.

We must tune the *actual value of* R, rather than choose a formal exponent independently. Turn on ordinary low dummy bits in cells from

$$\{4j : 0 \le j < N/5\} \cup \{1\}. \tag{65}$$

These cells do not wrap under either shift 1 or w_0. For a selected cell v, the exact changes are

$$\begin{aligned} \delta C &= \rho^{e_*} B^v, \\ \delta F &= (D_C + BD_R + B^{w_0})\rho^{e_*} B^v, \qquad \delta R = -\Gamma B^v, \end{aligned} \tag{66}$$

where

$$\Gamma = \rho^{e_*}(q^2-1)\big[1 + q(D_C + BD_R + B^{w_0})\big]. \tag{67}$$

Because b, L, N, w_0 are powers of five, $\rho, B, q, B^{w_0} \equiv 2 \pmod 5$. The bracket in (67) is $2D_C - D_R \equiv 3 \pmod 5$; hence Γ is a unit modulo dN.

For completeness, the Boolean subset lemma [13] says that every residue modulo dN is a sum of a subset of the distinct weights B^v with v from (65). Indeed B^4 has order $N/5$ modulo dN and its powers enumerate exactly the residues $1 + 5dt$. For a desired residue, optionally subtract the weight B to make its residue modulo 5 nonzero. Its residue k_0 modulo $5d$ then satisfies $1 \le k_0 < 5d < N/5$ and is a unit modulo $N/5$. A cyclic interval of k_0 distinct residues t can be chosen to have any desired sum, since its sum is $k_0 t_0 + k_0(k_0 - 1)/2$. This gives a genuine Boolean subset. The order assertion follows from $\nu_5(2^{4dj} - 1) = \nu_5(d) + 1 + \nu_5(j)$. Apply this lemma to the target $(R_0 - dw_0)\Gamma^{-1} \pmod{dN}$. The modified actual word and actual field give

$$R \equiv dw_0 \pmod{dN}. \tag{68}$$

The dummy bits are ignored by all clauses. Their upper bits remain zero, all masks remain valid, and the canonical pointwise digit bound gives

$$2C + F \le (\rho/4 + 1)\frac{q-1}{\rho - 1} < q/3. \tag{69}$$

Thus $J = (q-1)/(B-1) > 0$ and $\alpha = q - F - C > 0$. This verifies the stronger slack built into our definition of C, including after the dummy adjustment. By the exact shifted mask identity,

$$3q + 1 \le R < q^4, \qquad R \equiv 3 \pmod 4, \qquad \mathrm{pc}(R) = 3dN + 2. \tag{70}$$

For the congruence, (61) gives $C \equiv 1 \pmod 4$; the remainder T' in (58) is 3 (mod 4), and $S' \equiv 0$ (mod 4), so (58) gives $R \equiv 3 \pmod 4$.

Now invoke the positive half-binomial converse at this *fresh, actual* R. Set $r_0 = (R-1)/2$, $X = 2^R$, and

$$Y = \frac{1}{2}\left\lfloor \frac{(X+1)^{2r_0}}{X^{r_0}} \right\rfloor. \tag{71}$$

Equation (70) and the exact valuation imply $q^3 \mid Y$; also $q \mid X$. Let $s = Y/q^3$ and $w = X/q$. Both are positive integers. Set $a = Y(X+1)$, $A = a + 2$, $\Delta = A^2 - 1$, $E = XY$, $P_0 = 2XY^2 + 1$, and

$$\begin{aligned} &c = \psi_A(R), \quad D = \chi_A(R), \quad k = 2\psi_{P_0}(r_0 + 1), \quad \tau = \chi_{P_0}(r_0 + 1), \\ &\eta = c - kY, \quad \zeta = k - \eta, \quad h = (k - R - 1)/E, \quad \gamma = (D - X - ac)/(4a + 3). \end{aligned} \tag{72}$$

The kernel's strict ratio bounds and congruences make all these integers positive. The supplied quotient h here is different from the external helical width w_0. For the normalized strong auxiliary coordinates, let

$$m_1 = 2cR, \qquad f = \chi_A(m_1), \qquad i = \psi_A(m_1)/c^2, \qquad \rho_1 = \Delta\psi_A(m_1). \tag{73}$$

The quotient i is integral: the square-root coefficient of $(\chi_A(R) + c\sqrt{\Delta})^{2c}$ is divisible by c^2; its linear square-root term contains $2c^2$, and all higher odd terms contain at least c^3. The canonical minus construction [7, 9] at $R \equiv 3 \pmod 4$ gives

$$y = \psi_{\rho_1}(R), \quad V = \chi_{\rho_1}(R)/\rho_1, \quad o = (V + c)/f, \quad j = (V + R)/c \tag{74}$$

as positive integers satisfying the required norms. Since $of + R = c(j+1)$ and $f^2 \equiv 1 \pmod c$, $c \mid o + Rf$. Therefore

$$T = (o + Rf)/c > 0 \tag{75}$$

is integral and reconstructs exactly $V = c(Tf - 1) - Rf^2$. Finally (68) gives $X \equiv B^{w_0} \pmod{q-1}$, and the same holds for $w = X/q$. The actual field therefore makes $(K + w)C - F$ divisible by $q - 1$. In fact $w = 2^R/q > q > F$, by (70), so its quotient is positive. Put

$$z = 1 + \frac{(K + w)C - F}{q - 1} > 0. \tag{76}$$

Every equation in (8) now holds. Divisions and exponential formulas in this section construct existential witnesses in the proof; they are not operations in the delivered polynomial.

7 Why the equivalence is provable in ZFC

The pinned Lean endpoint [1] states, for the exact machine `machine215`,

```
blank_halts_iff_zfc_refutation :
  BlankHalts machine215 <-> BProv ZFCax_s [] Form.fBot

nonhalting_iff_zfc_consistency :
  (not (BlankHalts machine215)) <->
  (not (BProv ZFCax_s [] Form.fBot))
```

The Rocq endpoint `CorrectDense215.v` supplies `tm215_halts_iff_ZFC_inconsistent` and `tm215_nonhalting_iff_ZFC_consistent`. Here `BProv` means provability from some finite list of axioms in the given axiom predicate; its letter B does not impose a fixed proof-length bound. The underlying presentation has the ZF schemas and the stated choice principle. The ordinary calibration proof in the repository expressly gives the implication and converse as reasoning formalizable in ZFC [2]. Sections 2–6 give the mathematical equivalence

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{67}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Halt}(\mathrm{K215}, \mathrm{blank}). \tag{77}$$

Every definition is finite or an elementary recursively defined integer sequence. The proofs use induction, integer arithmetic, finite combinatorics, Pell theory, and explicit finite computation encodings. These constructions and arguments formalize in ZFC. Combining them with the calibrated K215 theorem proves Theorem 1.1. This does not infer an internal ZFC theorem merely from the existence of a Lean theorem: the mathematical argument supplies the formalizable proof, and the existing formalization is an independent calibration artifact.

Formal verification status. The archive records source identities and the existing endpoint statements. It does not contain a new Lean/Rocq proof of (77), nor a kernel-checked encoded ZFC derivation of (1). The repository's K215 verification dossier also distinguishes its endpoint theorems from a separately encoded internal ZFC or PA derivation. We inspected that dossier and the source endpoints; we did not rerun the entire Lean or Rocq kernel build in this work. The Python checks certify their stated finite properties, not the universal mathematical theorem.

Logical consequence. No consistency assumption is needed to prove the biconditional. To conclude that the equation actually has no solution, one must assume or establish Con(ZFC). Under the usual hypotheses for Gödel's second incompleteness theorem, if ZFC is consistent, ZFC cannot prove this absence of solutions. Mere consistency does not by itself justify the additional assertion that ZFC cannot prove the opposite Σ_1 statement. A suitable soundness assumption, such as Σ_1-soundness, suffices for that further conclusion.

8 Exact degrees and arithmetic counts

The pinned source `complete84_scaled_strong_output.json` yields the emitted 67-gate circuit by deleting the ordinary-input cone and its factor, removing the input-dependent split of γ, replacing the marked-word coordinate by the blank-word $C = q - F - \alpha$, and retaining the shared packing expression G. Its complete live ledger is $67 = 39M + 28A$. The relation to this parent is source provenance; soundness was proved above directly for the modified local language, not by erasing hypotheses from the parent's universal theorem. Give every unknown degree one and every fixed coefficient degree zero. Put

$$Q = mJ, \quad k_0 = \eta + \zeta, \quad C_1 = Q - F - \alpha, \quad T_0 = wC_1 - zQ. \tag{78}$$

Then

$$X_{\text{top}} = wQ, \quad Y_{\text{top}} = sQ^3, \quad a_{\text{top}} = wsQ^4, \quad c_{\text{top}} = k_0 sQ^3. \tag{79}$$

The exact factor degrees in (6) are

$$22, \quad 18, \quad 60, \quad 7, \quad 2, \quad 46. \tag{80}$$

The main norm has a significant cancellation:

$$N_1 = 2ac(X + \gamma H) + (X + \gamma H)^2 - Hc^2. \tag{81}$$

Its degree-18 leader is $8\gamma a_{\text{top}}^2 c_{\text{top}}$. The product's leading homogeneous form is

$$-8Q^{91} h\gamma i^4 k_0^{13} w^{13} s^{26} T^2 f^2 T_0. \tag{82}$$

For example, the degree-155 monomial

$$J^{92} h\gamma i^4 \eta^{13} w^{13} s^{26} T^2 f^2 z \tag{83}$$

has coefficient $8m^{92} \neq 0$. The final subtraction of Δ, of degree 12, cannot cancel it. The degree is therefore exactly 155, on this fixed coefficient slice. This calculation does not simplify using identities that hold only at solutions. For the reduced auxiliary coefficient,

$$P_{68} - P_{67} = (N_0 N_1 N_K N_T) N_S (N_S - \Delta)(V^2 - y^2). \tag{84}$$

At any zero of either polynomial with $\Delta \neq 0$, the integer unit N_s in (31) must be 1, since $\Delta \equiv 0, 3 \pmod 4$ excludes -1. Hence $N_S = \Delta$ and $S = \Delta(f^2 - 1)$; the two auxiliary factors agree. If $\Delta = 0$, both polynomials vanish identically because $N_S = 0$. Thus this deformation actually preserves their full integer zero sets as polynomials in the supplied coordinates, and in particular their required positive zero sets. The degree of N'_A is 28, and the new full leader is

$$-8Q^{71} h\gamma i^2 k_0^9 w^{11} s^{20} T^2 f^4 T_0. \tag{85}$$

Its $J^{72} h\gamma i^2 \eta^9 w^{11} s^{20} T^2 f^4 z$ coefficient is $8m^{72} \neq 0$, proving exact degree 123. One extra subtraction is needed, for $68 = 39M + 29A$. Deleting the five product-combination gates and final subtraction from (7) gives the 61-operation system. The analogous system for (9) costs 62. For a single lower-degree polynomial, sum the squares of the six residuals in (8). The six residual subtractions, six squares and five additions add 17 operations. The maximum residual degrees are 60 and 46; their unique highest-degree squares give exact degrees 120 and 92. The JSON system packets include both the comparison interface and these fully charged sum-of-squares finalizers.

9 An equation over all integers

The phrase *Diophantine equation* often prescribes variables in $\mathbb{Z}$, rather than $\mathbb{Z}_{>0}$. The following explicit conversion resolves that domain issue without division.

Lemma 9.1 (Positive integers with three signed coordinates)**.**

$$\mathbb{Z}_{>0} = \{1 + u^2 + v^2 + r(r+1) : u, v, r \in \mathbb{Z}\}. \tag{86}$$

Proof. The expression is positive because $r(r+1) \geq 0$. Conversely let $n \geq 1$. The number $4(n-1)+1$ is $1 \pmod 4$ and is not of the excluded form $4^a(8b+7)$. By Legendre's three-square theorem it is a sum of three integer squares. Exactly one summand has an odd root. Write the roots, after reordering, as $2u, 2v, 2r+1$ and divide the equality by four. This proves (86). The same integer parametrization has a formalized proof in the Archive of Formal Proofs [14]. □

The coordinates τ, y occur only through their squares. With the other thirteen coordinates positive, neither can be zero at a zero of P_{67} or P_{68}. Indeed $\tau = 0$ would make $N_0 < -1$; after the strong unit is recovered, $y = 0$ would make the auxiliary unit $SV^2 = 1$ with $S > 1$, impossible. Allowing them to be signed therefore changes no solvability statement: take absolute values. For each of the thirteen coordinates in

$$\mathcal{I} = \{J, F, \alpha, z, f, h, i, T, s, w, \eta, \zeta, \gamma\}, \tag{87}$$

introduce three separate integer unknowns u_x, v_x, r_x, and set

$$x = 1 + u_x^2 + v_x^2 + r_x(r_x + 1). \tag{88}$$

Keep $\tau, y \in \mathbb{Z}$. Equations (4)–(7), (25), and (88) now specify one ordinary polynomial in $13 \cdot 3 + 2 = 41$ unrestricted integer variables. Its absence of integer solutions is provably equivalent in ZFC to Con(ZFC). The packets with suffix `over_Z41` spell out every gate. Each substitution costs seven operations, for total costs $67 + 13 \cdot 7 = 158$ and $68 + 13 \cdot 7 = 159$. The exact degrees double to 310 and 246: the leading forms (82) and (85) contain neither τ nor y, and replacing the other variables by the disjoint quadratic forms $u_x^2 + v_x^2 + r_x^2$ is injective on the polynomial ring. Set $v_x = r_x = 0$ to see the injective monomial map $x \mapsto u_x^2$.

10 Reproduction, scope, and further work

10.1 What the archive contains

The main files are `report.tex` and `report.pdf`. The `data/` directory contains the literal polynomial and system circuits, the full transition list, the finite-expression coefficient recipe, independent cardinality and local-semantics receipts, and source manifests. The `code/` directory contains standard-library Python emitters and auditors. The `proofs/` directory supplies additional readable arithmetic, tableau, and source-calibration notes. The `upstream/` directory preserves a scoped set of proof dependencies and calibration sources from the pinned repository revision, with original relative paths. The applicable upstream licenses are in `licenses/`. From the extracted archive, run:

```
python3 -B code/verify_all.py --output verification_receipt.json
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Python needs no third-party packages. The audit uses a temporary directory and compares rebuilt receipts with the distributed data. No network connection, historical research script execution, actual expansion of B, or actual halting witness is needed. The root `README.md` gives the supported options and the final recorded result.

10.2 Evidence and its limits

The finite audits establish:

- exact agreement of all 430 transition rows in the three exported K215 representations and the stated hashes;
- agreement of the local predicate with finite halting controls, independent padding and helical seam checks, and rejection of a nonhalting truncation;
- the exact window cardinality, by closed category counts and a separate direct predicate enumeration;

- deterministic reconstruction of all arithmetic circuits, live ports and gates, operation ledgers, independent exact evaluations, and degree diagnostics supported by the uniform leading-form proof;
- the scalar constant recipe, its no-carry bounds and modulo-five identity, with independent sparse small-layout checks of band and anchor geometry.

They do not decide whether K215 halts, produce a ZFC refutation, expand the giant fixed coefficients, search for an actual solution, or replace the universal mathematical arguments above. The proof is intended to be reviewable and suitable for subsequent formalization; the finite checks should not be described as a kernel certificate of the final consistency equivalence.

10.3 Useful next improvements

The most valuable next formal step is to encode the exact local predicate, native-word soundness, and arithmetic rank composition in Lean or Rocq and connect them to the existing K215 endpoint. This would turn the distinction just stated into a machine-checked bridge. For mathematical simplification there are several different objectives. A direct compiler for the overlapping cross relation could reduce coefficient-description cost substantially by avoiding the full 3-by-3 window alphabet. It would require a fresh carry-free and synchronization proof; such a reduction is not claimed here. Applying an effective nine-variable reduction to this explicit polynomial is another concrete project, but its complete specialized formula and coefficient recipe must actually be emitted before claiming that alternative. Reducing the remaining positivity witnesses or changing the Pell scales requires new full-zero-set arguments, since deleting a coordinate based only on canonical witnesses can create spurious solutions. No global simplest Diophantine equation is asserted without a chosen ordering of variable count, degree, arithmetic cost, coefficient height, and proof complexity. The archive makes these tradeoffs measurable for one fully specified consistency instance.

A Complete arithmetic circuit

Every row is one binary operation. The two fixed-numeral port names that need care are `Bm1` for m and `MF` for M_F^{src}. The source register `A` denotes Δ, not the Pell parameter $a+2$. The input aliases are `Jrep`=J, `transport_quotient`=z, `auxiliary_quotient`=T, `tau_root`=τ, and `y_aux`=y.

No.	Output register	Operation
1	`tau_square`	`tau_root * tau_root`
2	`repunit`	`Bm1 * Jrep`
3	`q`	`repunit + 1`
4	`Lbig`	`q * q`
5	`n2`	`Lbig * q`
6	`wn2`	`w * q`
7	`sn2`	`s * n2`
8	`UM`	`wn2 * sn2`
9	`R10b`	`eta + zeta`
10	`ksn2`	`R10b * sn2`
11	`first_root_base`	`UM * ksn2`
12	`first_next`	`first_root_base + R10b`
13	`first_product`	`first_root_base * first_next`
14	`norm_first`	`tau_square - first_product`
15	`R10a`	`ksn2 + eta`
16	`R12`	`UM + sn2`
17	`cam2`	`R10a * R12`
18	`D1`	`wn2 + cam2`

No.	Output register	Operation
19	a4	4 * R12
20	a4m5	a4 + 3
21	gam	gamma * a4m5
22	R14	D1 + gam
23	L15	R14 * R14
24	a_square	R12 * R12
25	A	a_square + a4m5
26	c2	R10a * R10a
27	Ac2	A * c2
28	norm_main	L15 - Ac2
29	norm_pair	norm_first * norm_main
30	q_minus_F	q - F
31	C	q_minus_F - alpha
32	aux_y2	y_aux * y_aux
33	hpm1	h * UM
34	index_difference	R10b - hpm1
35	gap_product	repunit * q_minus_F
36	gap	gap_product + alpha
37	Lm1	Lbig - 1
38	rproduct	gap * Lm1
39	qMF	q * MF
40	mask_factor	MC + qMF
41	mask	mask_factor * Jrep
42	r_lhs	rproduct + mask
43	norm_index	index_difference - r_lhs
44	kinner	Kconstant + w
45	innerC	kinner * C
46	transport_partial	innerC + q_minus_F
47	local_rhs	transport_quotient * repunit
48	norm_transport	transport_partial - local_rhs
49	L16	f * f
50	auxiliary_Tf	auxiliary_quotient * f
51	auxiliary_Tf_minus_one	auxiliary_Tf - 1
52	auxiliary_c_Tf	R10a * auxiliary_Tf_minus_one
53	auxiliary_R_f2	r_lhs * L16
54	aux_u_rhs	auxiliary_c_Tf - auxiliary_R_f2
55	H2	aux_u_rhs * aux_u_rhs
56	aux_square_gap	H2 - aux_y2
57	aux_coefficient_root	i * Ac2
58	R16	aux_coefficient_root * aux_coefficient_root
59	scaled_f_square	A * L16
60	norm_strong	scaled_f_square - R16
61	L17	R16 * aux_square_gap
62	norm_aux	L17 + aux_y2
63	norm_four	norm_pair * norm_aux
64	norm_product	norm_four * norm_index
65	all_units	norm_product * norm_transport
66	seven_units	all_units * norm_strong
67	polynomial	seven_units - A

B Independent local-window cardinality

Only the five-position cross enters the local predicate. The four corners contribute the free factor 439^4. The category table below records the number of valid crosses per center and the number of center symbols in that category. The direct checker sums over every left/right pair for a representative of each category and independently computes its allowed previous/next combinations. The grouping of moving head states is valid because the new state of a moving center head does not occur in that center's own next payload.

Center category	Multiplicity	Crosses per center
Vertical boundary	1	84,411,798
Horizontal boundary	1	168,438,154
Headless, symbol 0	1	184,091,400
Halt, symbol 0	1	98,097,108
Headless, symbol 1	1	155,712,504
Halt, symbol 1	1	98,097,108
Nonhalting, write 0, left	51	99,368,184
Nonhalting, write 0, right	190	99,600,324
Nonhalting, write 1, left	31	71,301,144
Nonhalting, write 1, right	158	71,301,144
Initial L	1	2,118
Initial Q	1	437
Initial R	1	2,110
All centers	439	Total: 38,256,607,897

C Provenance and proof-dependency map

The source snapshot is

`cdd582bec13f908df3711fa9e05d1ddac368a5db`.

The immediate arithmetic parent has SHA-256

`8b2bc5d0b4db90c168107b7ded2cb4ca08df0098ed190851a3db4f1e49a9c0cf`.

The machine hash appears in Section 3. The machine/source agreement manifest and the complete archive hash manifest are separate from the polynomial's mathematical coefficients: hashes identify the supplied bytes, while (10)–(25) define the mathematical objects. The imported proof collection is deliberately scoped. Historical progress headers may point to other research packets; those headers do not enlarge the dependency claims of this paper. The detailed machine-readable `data/proof_dependencies.json` records the imported sections and file hashes.

Obligation	Principal included proofs
Strong rank and signed target	Generic relaxed auxiliary rank, signed Pell proof, half-parameter Pell proof, reversed auxiliary and exact quotient projection
Power and binomial extraction	Base-two Pell proof, half-binomial kernel, asymmetric scale proof
Normalized strong converse	Normalized strong auxiliary witness; explicit divisibility at $m_1 = 2cR$
Native masks and alignment	Fixed native compiler 76; modified half-binomial compiler; canonical bounded-projection margin
Actual-index tuning	Five-adic Boolean dummy-control theorem
Blank fixed-input semantics	New `blank_helical_proof.md` and exact executable predicate
ZFC calibration	K215 ordinary proof, Lean endpoint, Rocq endpoint, verification README

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